Geometry : Vectors — problem 12.1.19

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In a tetrahedron $OABC$ with edges of lengths $|OA| = |BC| = a$, $|OB| = |AC| = b$, $|OC| = |AB| = c$, let A_1 and C_1 be the centroids of the triangles ABC and AOC, respectively. Prove that, if $OA_1 \perp BC_1$, then $a^2 + c^2 = 3b^2$.